Lohit Gali

Senior Backend Payment Services Engineer

lohitgali@gmail.com | +1(475)-291-1455 | Seattle, WA

PROFESSIONAL SUMMARY

Senior Backend Payment Services Engineer with 5+ years of experience designing, building, and maintaining scalable backend payment services for global platforms. Proficient in Java/J2EE and Kotlin, specializing in secure data handling within cloud-based and distributed systems. Expertise in CI/CD, architectural patterns, and leveraging GenAI technologies to enhance system reliability and innovation.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Python, SQL, JavaScript

Cloud Platforms & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines, Terraform, Observability, Automation

Databases & Data Management: Relational Databases (PostgreSQL, MySQL), NoSQL Databases (MongoDB, DynamoDB), Distributed Systems, Data Structures

Architecture & Methodologies: Microservices Architecture, Software Design Principles, Design Patterns, Agile Development, Scrum, System Design, Root-Cause Analysis

AI & GenAI Technologies: GenAI Capabilities, LLMs, Prompt Engineering, Tool-Calling Agents

PROFESSIONAL EXPERIENCE

Lead Backend Services Engineer | Epik Solutions

Jul 2025 - Present

• Designed and implemented highly scalable backend payment services utilizing Java/Kotlin, cloud-based distributed systems (AWS/Azure), and advanced software design principles to support Expedia’s global travel platform; improved transaction throughput by 40% and reduced latency by 25% for critical payment flows.

• Architected secure data handling practices and robust data structures for sensitive payment information within a microservices architecture, integrating with relational and NoSQL databases; ensured compliance standards (e.g., PCI DSS) achieving 100% compliance audit success rates and preventing zero data breaches.

• Led the enhancement of service reliability through comprehensive observability and automation strategies, implementing monitoring, logging, and incident resolution protocols across complex distributed systems in production; reduced critical incident resolution time by 30% and improved system uptime to 99.99%.

• Collaborated with cross-functional teams (product, frontend, finance) utilizing Agile development environments and CI/CD pipelines to deliver high-impact payment features from design to deployment; accelerated feature delivery by 20% and ensured alignment with Expedia’s strategic goals.

• Recommended and applied innovative architectural patterns and GenAI capabilities to optimize system design and drive innovation in payment fraud detection and customer service automation; integrated intelligent agents for real-time anomaly analysis, reducing manual fraud review by 15%.

• Drove continuous operational improvements by conducting root-cause analysis on performance bottlenecks and implementing modular, optimized code to enhance maintainability and scalability of backend services across multi-quarter, cross-team projects; decreased operational costs by 10% and improved developer velocity by 20%.

• Mentored junior engineers on best practices in software design principles, secure coding, and cloud deployment strategies, fostering a culture of technical excellence and continuous learning; improved team code quality by 25% and reduced onboarding time for new hires by 15%.

Senior Backend Engineer | Samay Systems LLC

Jul 2024 - Jun 2025

• Developed robust backend services using Java/J2EE and object-oriented languages, integrating with various relational and NoSQL databases for high-volume data processing to support a compliance platform SaaS; processed over 1 million transactions daily with 99.9% data integrity, enhancing platform reliability.

• Implemented scalable APIs and microservices leveraging a cloud environment (AWS, Azure) for secure data collection and validation; automated compliance workflows and evidence gathering from diverse sources, reducing

manual effort in compliance audits by 35% and accelerating reporting cycles by 20%.

• Enhanced CI/CD pipelines and infrastructure automation using Jenkins and Terraform, ensuring rapid and reliable deployment of backend features in Agile development environments; decreased deployment time by 50% and reduced production errors by 15%, improving development team efficiency.

• Contributed to architectural discussions and system design, focusing on scalability, performance, and security for cloud-based distributed systems at scale; designed a new data ingestion pattern that scaled data throughput by 2x, preparing the platform for anticipated traffic spikes and GenAI technologies.

• Provided production support including monitoring, debugging, incident resolution, and root-cause analysis for complex distributed systems to maintain high service availability and operational excellence; achieved a 99.98% uptime SLA and resolved 95% of critical incidents within target MTTR.

Backend Software Engineer | Arvam Tech

Jul 2023 - May 2024

• Developed and optimized backend data ingestion and transformation pipelines using Python and SQL for an Enterprise Asset Management System, consolidating operational datasets into analytics-ready formats; improved data processing efficiency by 30% and ensured data availability for trend analysis 24/7.

• Implemented event-driven architectures for processing streaming data, ensuring near-real-time availability of analytical data in a cloud environment (Azure) to support dynamic updates and operational intelligence; decreased data staleness by 80% and supported faster decision-making for asset utilization.

• Designed and deployed scalable services for reporting and metrics frameworks, utilizing software design principles and clean, maintainable code to surface utilization patterns and operational risks for decision-makers; enhanced risk identification accuracy by 20% and reduced reporting generation time by 40%.

• Automated backend processes and integrations using scripting and API development to sync asset inventory and process webhook-driven events, enhancing operational efficiency; eliminated 15 hours of manual data entry per week, reallocating resources to strategic tasks.

• Collaborated on improving backend reliability and performance of internal tools through diligent debugging and optimization of database queries and service calls; decreased system downtime by 10% and enhanced user satisfaction ratings by 15%.

Associate Backend Engineer | RHYM Tech

Aug 2022 - May 2023

• Developed backend services for a Cybersecurity Threat Intelligence Platform using Python and SQL to analyze high-volume telemetry and event data; processed over 500 GB of daily event data, enabling real-time threat detection and supporting statistical analysis.

• Implemented data cleaning, transformation, and feature extraction processes, ensuring data quality for downstream machine learning tasks and analytical dashboards; improved data accuracy by 25% and reduced data processing errors by 10%.

• Built Python-based data pipelines to ingest high-volume event data into structured stores, enabling robust statistical analysis and anomaly detection to identify emerging trends across security telemetry; detected critical security anomalies 12 hours faster than previous methods.

• Applied time-series aggregation and anomaly detection logic within backend services to identify emerging trends across security telemetry data, enhancing proactive threat intelligence capabilities; reduced false positive alerts by 15% through optimized detection algorithms.

• Collaborated on visualizing data distributions, trends, and risk indicators, supporting both technical and non-technical stakeholders with clear analytical dashboards; enhanced understanding of security posture, leading to a 5% reduction in overall system vulnerabilities.

Associate Software Developer | ZeMoSo Technologies

Jul 2021 - Aug 2022

• Developed backend APIs and data access layers for a Biotech LabOps Automation project using object-oriented programming principles and secure coding practices to support UI-driven workflows and ensure data integrity; improved data retrieval efficiency by 20% and supported an average of 500 daily lab operations without downtime.

• Implemented RESTful APIs and integrated with relational databases, balancing correctness, performance, and maintainability for critical backend services; ensured 99.9% API uptime, crucial for continuous lab operations and data exchange.

• Assisted in building analytical pipelines and reports aligned with business requirements through data preparation and exploratory analysis, supporting data science initiatives; enabled faster generation of 10+ monthly analytical reports, saving 5 hours of manual work per report.

• Contributed to the development of backend components, adhering to computer science fundamentals and modular design principles, to enhance the overall system architecture and scalability; reduced code complexity by 10%, facilitating easier future enhancements.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India